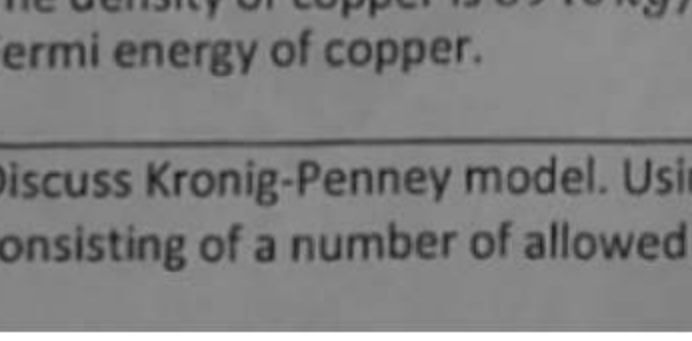
MID-TERM EXAMINATION
(Course Name:B.Tech IT/MAE/DMAM) (Semester: Second)
(May, 2023) OFF LINE mode

Subject Code: BAS-104	Subject: APPLIED PHYSICS-II
Time: 1 ½Hours	Maximum Marks: 30
Note:Q. 1 is compulsory.	

Q1		(2.5*4)	
	(a) Explain Poynting's Theorem. Discuss the physical significance of the Poynting vector.		
	(b) Calculate the skin depth for silver if the frequency ofincident EM wave is given as10^8THz and $\sigma = 6 \times 10^7 S/m$. Use: $\mu_r \approx 1$ and $\mu_o = 4\pi \times 10^{-7} N/A^2$		
	(c) A Bravais lattice is specified by the primitive lattice vectors: $\vec{a} = a\,\hat{x}, \quad \vec{b} = \frac{1}{2}a\,\hat{x} + \frac{\sqrt{3}}{2}a\,\hat{y}, \quad \vec{c} = 2a\,\hat{k}$ Where $\hat{x}$, $\hat{y}$ and $\hat{z}$are the unit vectors along the x, y, and z axes respectively, and a is the lattice distance. Draw the Wigner-Seitz primitive cell for the lattice in the $x-y$plane.		
	(d) Assuming a simple cubic lattice with lattice constant $a = 1$. Draw the crystal planes corresponding to the **$(0\bar{2}1)$**, **(123)**and **$(20\bar{2})$** Miller indices?		
Q2	(Attempt any Two Parts) UNIT-1		(5,5)
	(a) Discuss the propagation of EM waves in a conductor. Briefly describe why we get very low mobile signal in a lift?		
	(b) Prove that in free space, the electromagnetic waves are transverse waves.		
	(c) A parallel plate air-gap capacitor is made up of two plates of area $10cm^2$each kept at adistance of $0.88mm$. A sine wave of amplitude $10V$ and frequency $50Hz$ is applied across the capacitor as shown below. Find the amplitude of the displacement current density. 		
Q3	(Attempt any Two Parts) UNIT-2		(5,5)
	(a) What are Fermions and Bosons? Describe and Sketch their respective distribution functions for temperatures $T_1 = 0K$ and$T_2 > T_1$. Using the Fermi-Dirac Distribution function, explain the concept of Fermi Energy.		
	(b) The density of copper is $8940\,kg/m^3$ and atomic weight is $63.55amu$. Determine the Fermi energy of copper.		
	(c) Discuss Kronig-Penney model. Using the model show the energy spectrum of electrons consisting of a number of allowed energy bands separated by forbidden bands.		